Probabilistic Forecasting of ICE Coffee C Futures Returns via GARCH(1,1) with Student's t Innovations

Bridget Brinkman, Faith Morrisey, Yun-Chih (Rick) Wu, Stella McElwain, and Farah Hendawy

***Abstract*—This paper develops a probabilistic forecasting model for daily ICE Coffee C front-month futures, motivated by the procurement-planning needs of a Pittsburgh-area specialty coffee roaster. We proceed in two parts. Section I constructs a 60-origin rolling-window backtest of ten point-forecast models, ranging from a naïve drift baseline to the IBM Granite Tiny Time Mixer foundation model. Pairwise Diebold-Mariano tests and a Model Confidence Set find that nine of the ten models are jointly statistically indistinguishable from random-walk-with-drift at $\alpha = 0.10$ on mean absolute error. Section II argues that this is not evidence of an unstructured series but of an inappropriate forecasting target. Coffee returns exhibit pronounced volatility clustering and fat tails; these are exploitable by density forecasts even when point forecasts cannot be improved. We fit a GARCH(1,1) model with standardized Student's t innovations and evaluate its calibration on the same 60-origin walk-forward backtest. Empirical coverage of the model's prediction intervals tracks nominal coverage within a few percentage points at every level evaluated, and predictive-interval widths adapt smoothly to local volatility across diverse historical regimes.**

***Index Terms*—Time series, GARCH, density forecasting, volatility clustering, coffee futures, probabilistic forecasting, interval calibration, Model Confidence Set.**

I. Rolling-Window Backtest of Point Forecasts

A. Background and Motivation

THIS project was motivated by the procurement-planning needs of 19 Coffee Company, a wholesale specialty coffee roaster based in the South Hills of Pittsburgh, Pennsylvania. 19 Coffee sources beans through direct-trade relationships with smallholder farms in regions including Ethiopia, Colombia, Guatemala, Sumatra, and Kenya, and supplies cafés, gourmet groceries, and restaurants across the Pittsburgh tri-state area. While 19 Coffee does not transact directly in the futures market, the ICE Coffee C contract is the global price benchmark to which their input costs are ultimately tied through importer pricing and the differential-to-C system (specialty contracts are priced as a fixed premium above the C futures benchmark, so 19 Coffee's procurement costs track C movements without direct market participation). A forecasting framework that produces calibrated uncertainty intervals—rather than single-number point estimates—would inform their procurement timing, contract negotiation, and budget planning. Our objective across both sections of this paper is to develop and validate such a framework.

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B. Data

We use the daily front-month nearby continuous settlement series of ICE Coffee C futures, obtained from the ICE Report 293 historical price service.[1] The series spans January 1994 through January 2026 ($N = 8{,}104$ daily observations, $8{,}103$ daily log returns after differencing). Prices are quoted in cents per pound (¢/lb). Holiday closures and exchange suspensions appear as gaps in the trading-day index but introduce no missing values in the cleaned series.

C. Forecasting Model Suite

We select ten forecasting models spanning a deliberately wide range of modeling paradigms—from naïve baselines through classical statistical methods to recent foundation models—with the intent of testing whether any class of model is able to extract structure that the others miss. All models are evaluated through a uniform `predict(context_df, horizon)` interface so that the backtest runner treats them as interchangeable.

1) Naïve baselines (from `statsforecast` [1]): **Naïve** repeats the last observed price for all horizons. **RWD** (random walk with drift) repeats the last observed price plus a linear drift equal to the mean training-window return.

2) Classical statistical models (from `statsforecast`): **Theta** decomposes the series into long-term trend and short-term components. **AutoETS** selects an exponential-smoothing specification (error, trend, seasonality combinations) by information criterion. **AutoARIMA** selects an ARIMA(p, d, q) specification similarly.

3) Recursive machine-learning models: **Ridge** (linear regression with ℓ_2 penalty) and **Random Forest** [2] are fit on lag-based features (lags $1, 2, 5, 10, 21, 63$) and rolling-window summaries (mean and standard deviation over $5, 21, 63$ days), then iterated forward recursively by feeding each prediction back as the next observation.

4) Volatility model: **GARCH** uses an AR(1) mean equation with GARCH(1,1) variance and Gaussian innovations, fit on raw prices, with the point forecast taken from the mean equation's h-step prediction. We note that this is a *different* parameterization from the GARCH(1,1)+t model used in Section II, where the mean equation is fixed at zero, the model is fit on log returns, and the innovations are Student's t. The model in this section is used as a point forecaster for direct

[1]https://www.ice.com/report/293

comparison with other point-forecast methods; Section II uses GARCH as a density forecaster.

5) Decomposition model: **Prophet** [3] with yearly seasonality and no daily seasonality.

6) Foundation model (zero-shot): **Granite TTM** [4] is the IBM Granite Tiny Time Mixer R2, a transformer-based foundation model for time-series forecasting pretrained on a large multi-domain corpus. We use the zero-shot variant with a 1 536-step context window; the context is per-window standard-scaled before inference and de-scaled after. Granite's context length requirement is what motivates the 1 536-day training window used uniformly across all models in this backtest.

D. Backtest Design

We use a fixed-length rolling-window backtest. At each forecast origin T_k, models are fit on the prior 1 536 trading days $\{P_{T_k-1535}, \ldots, P_{T_k}\}$ and forecast 63 trading days ahead (approximately one calendar quarter). Forecast accuracy is evaluated against the realized prices $\{P_{T_k+1}, \ldots, P_{T_k+63}\}$.

Choice of number of origins. The number of origins evolved during the project. At one origin (the most recent 63 days of data), Granite-TTM produced the lowest mean absolute error (20.88 ¢/lb versus RWD's 29.23, a 28 % gap), consistent with our initial expectation that mechanistic complexity would correlate with forecast accuracy. Fig. 1 shows all ten models' forecasts at this single origin; most predict a flat-to-upward path while the realized price declined sharply, and even the three models that trended downward (Granite-TTM, RF, GARCH) substantially underestimated the magnitude of the decline. We then scaled to 10, 30, and finally 60 evenly-spaced origins, with the spacing fixed at 110 days such that consecutive forecast horizons do not overlap. At 60 origins, the design covers approximately 80 % of the full series (∼2000 through 2025). Fig. 2 visualizes the four origin configurations on the price timeline. The rankings at 60 origins, reported below, *invert* the single-origin verdict: Granite-TTM falls to eighth (MAE 13.84) and RWD becomes the leader (MAE 12.79).

Evaluation metrics. We report mean absolute error (MAE) as the primary metric, with root mean squared error (RMSE) and symmetric mean absolute percentage error (sMAPE) reported as supporting metrics.

E. Leaderboard

Table I reports mean error metrics across all 60 forecast origins, sorted ascending by MAE.

Three features of the leaderboard warrant comment. First, RWD produces the lowest mean MAE despite being one of the simplest models in the suite. Second, the top four models (RWD, ETS, Theta, Naïve) sit within 0.12 ¢/lb of each other—a gap small enough to fall well within any reasonable margin of sampling noise. Third, Granite-TTM, the foundation model whose pretraining was the source of our initial optimism, places eighth out of ten and is outperformed by every classical statistical baseline. Prophet sits well outside the cluster of nine and forms the only large absolute gap on the leaderboard.

TABLE I
POINT-FORECAST LEADERBOARD AT 60 ROLLING FORECAST ORIGINS, SORTED ASCENDING BY MEAN MAE (CENTS/LB).

Model	MAE	RMSE	sMAPE
RWD	12.79	14.84	0.083
ETS	12.82	14.89	0.085
Theta	12.86	14.93	0.085
Naïve	12.91	14.99	0.085
RF	13.13	15.19	0.087
ARIMA	13.24	15.39	0.085
GARCH	13.50	15.74	0.090
Granite-TTM	13.84	16.09	0.092
Ridge	14.68	17.16	0.101
Prophet	22.69	24.99	0.152

F. Statistical Tests: Diebold-Mariano and the Model Confidence Set

The leaderboard is descriptive but does not resolve statistical significance. We apply two complementary tools.

1) Diebold-Mariano test: Each model is compared pairwise to the RWD baseline using the Diebold-Mariano test [5], with the Harvey-Leybourne-Newbold small-sample correction [6]. The loss series is per-origin MAE (60 values per model), and the long-run variance is estimated with a Newey-West HAC estimator [7] using the Bartlett kernel and the rule-of-thumb bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$.

2) Model Confidence Set: We apply the Model Confidence Set (MCS) of Hansen, Lunde, and Nason [8] to identify the set of models *jointly* indistinguishable from the best, at significance level $\alpha = 0.10$. Null distributions are obtained by circular block bootstrap with block size 5 and $B = 5{,}000$ replicates.

3) Results: Fig. 3 summarizes both tests.

The pairwise DM test rejects the equal-accuracy null at $\alpha = 0.05$ for Granite-TTM ($p = 0.022$), Ridge ($p = 0.046$), and Prophet ($p < 0.001$), and marginally for GARCH ($p = 0.083$). All four top-tier simple models (ETS, Theta, Naïve, RF) have $p > 0.4$ against RWD.

The MCS at $\alpha = 0.10$ retains nine of the ten models, eliminating only Prophet (MCS $p = 0.009$). The retained set spans every category in the model suite: naïve baselines, classical statistical models, recursive ML models, the volatility model, and the foundation model.

The MCS result is the appropriate primary inference because it incorporates the multiple-comparisons correction implicit in evaluating ten models against a single baseline. The pairwise DM rejections of Granite-TTM and Ridge do not survive joint correction; with ten candidates being compared, the family-wise Type I rate inflates well above the nominal α of any single test. We note also that $T = 60$ origins is small relative to the simulation regimes in which the MCS test is typically calibrated ($T \geq 100$), so the wide MCS partly reflects limited test power; further origins would likely shrink the surviving set, but the qualitative conclusion—no mechanism in this suite reliably beats RWD—is robust.

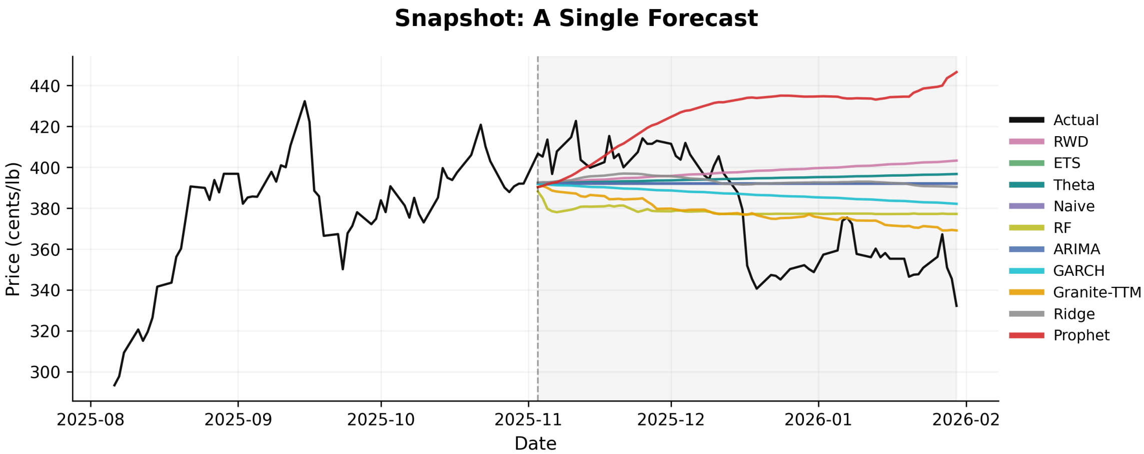

Fig. 1. Forecasts from all ten models at the single most recent forecast origin (November 2025). The dashed vertical line marks the origin; historical prices appear to its left and 63-day forecasts from all ten models to its right. The actual price declined from approximately 400 to 330 ¢/lb over the horizon. Naive, ETS, and ARIMA produce nearly identical flat-line forecasts at this origin and overlap heavily in the plot.

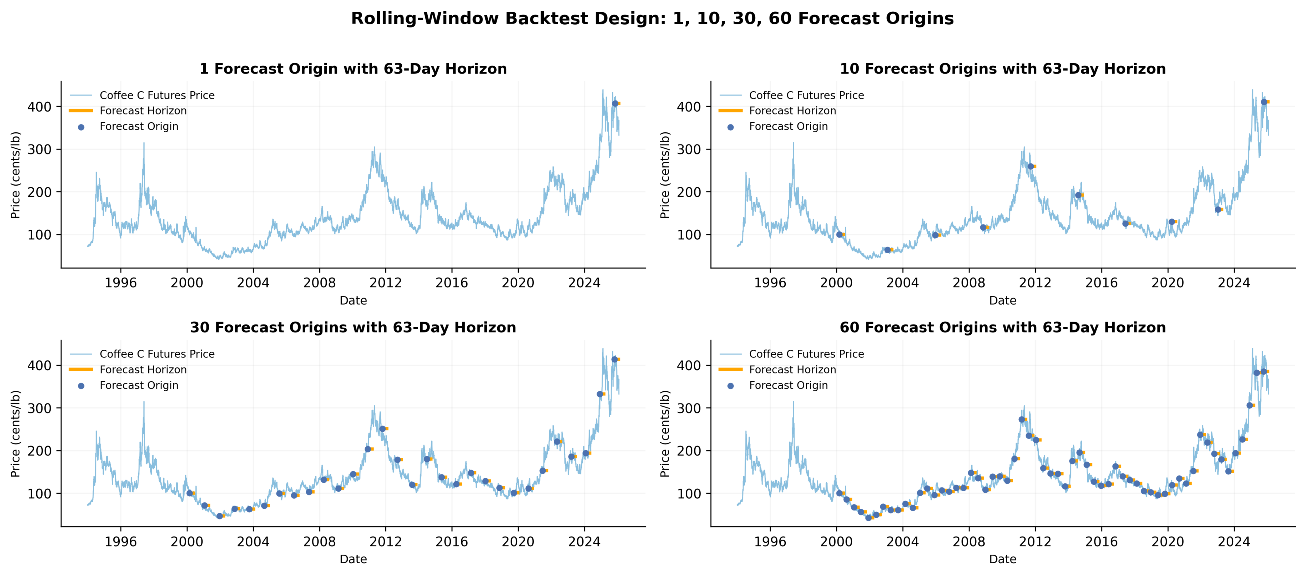

Fig. 2. Rolling-window backtest design at four scales: 1, 10, 30, and 60 forecast origins. Each blue dot marks a forecast origin; each orange segment shows the 63-day forecast horizon evaluated against realized prices. The single-origin design (top left) places its only test at the end of the series, during the 2024–2025 rally—conditions that, by chance, favor mechanically complex models. The 60-origin design (bottom right) covers approximately 80 % of the available series and samples many regimes; this is the design under which the leaderboard in Table I is computed.

G. Section I Conclusion: The Random-Walk Boundary

A diverse ten-model suite, ranging from naïve baselines to a foundation model with cross-domain pretraining, cannot be statistically distinguished from the random-walk-with-drift forecaster on the coffee front-month series at 60 origins. This pattern admits a natural interpretation in financial economics: it is consistent with weak-form market efficiency [9], the hypothesis that past prices alone contain no exploitable information about future prices, and is the empirical fingerprint one would expect on a price series whose direction is essentially unpredictable from its own history.

This finding could be read narrowly as “no model helps,” but that reading would mistake the scope of the test. Section I evaluates *point* forecasts under *absolute-error* loss. Coffee returns contain substantial structure beyond their mean—in particular, time-varying volatility and fat tails—that point-forecast MAE is constitutionally incapable of rewarding. Section II reframes the forecasting target to expose this structure.

II. Probabilistic Forecasting via GARCH(1,1)

A. From Point to Density Forecasts

A point forecast collapses a predictive distribution into a single number, typically its mean or median. For a series whose direction is approximately unpredictable on short horizons, the best point forecast of tomorrow’s price is essentially today’s price, and the best point forecast over a horizon h is the current price (or current price compounded by a tiny drift). This is what Section I observed: across ten candidate models, no method materially outperformed RWD on MAE.

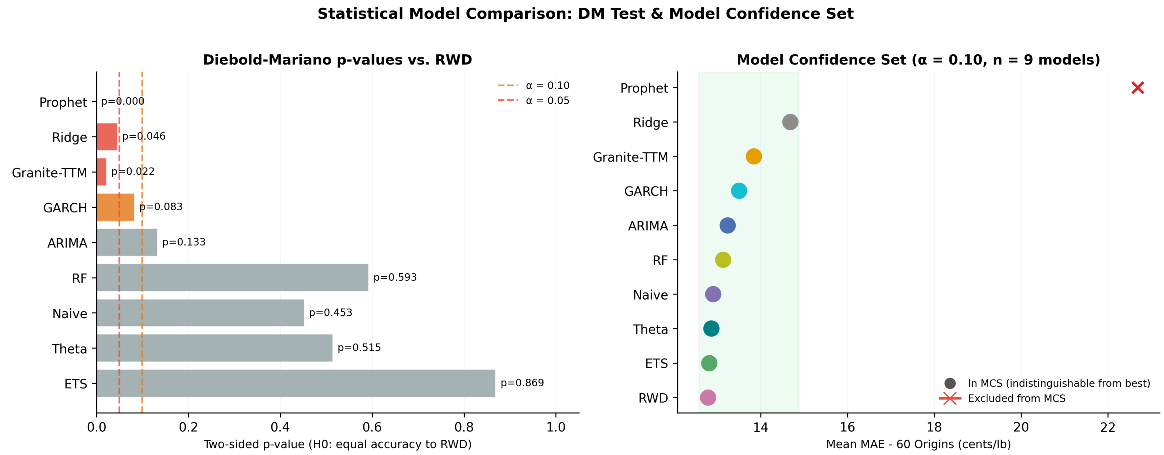

Fig. 3. Statistical comparison of the ten point-forecast models at 60 origins. (a) Pairwise Diebold-Mariano p-values testing each model against RWD; bars are colored red for significance at $\alpha = 0.05$ and orange at $\alpha = 0.10$. (b) Model Confidence Set membership at $\alpha = 0.10$; filled dots are retained, the red cross is excluded. Nine of ten models—all except Prophet—are jointly indistinguishable from the best.

Point-forecast MAE, however, ignores the spread of the predictive distribution. A point forecast tells a procurement manager what the "expected" price is; a density forecast tells them what range of prices is plausible. An interval forecast can be used to size positions, price options, set risk limits, or trigger contingent decisions, in ways that a point forecast cannot. This is the forecast object 19 Coffee would actually use—an interval rather than a point—and it is the object Section II constructs.

The structure that motivates this richer forecasting target is well known in the empirical finance literature [10]–[12] and is clearly visible in the coffee series. We summarize the relevant *stylized facts* below.

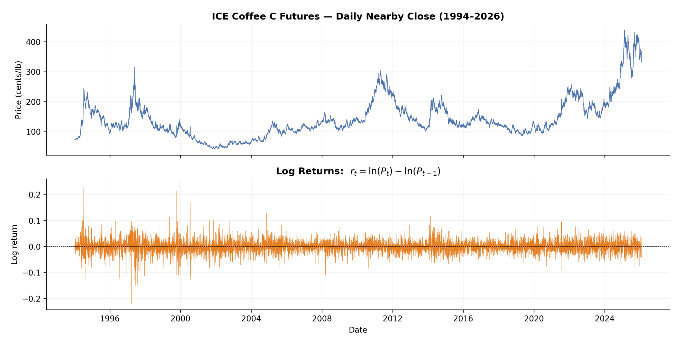

Fig. 4. ICE Coffee C nearby front-month futures. (a) Daily settlement price, January 1994 through January 2026. (b) Corresponding daily log returns. Note the visibly higher amplitude of returns during 1994–2000 and 2024–2025 compared to 2005–2010 or 2017–2020.

B. Data and Stylized Facts

Throughout, P_t denotes the daily settlement price (cents per pound) at trading day t, and $r_t = \log(P_t/P_{t-1})$ denotes the daily log return. Log returns are preferred over simple returns because they are additive across time: the h-day log return is the sum of h consecutive daily log returns.

Fig. 4 displays P_t together with r_t. Three features are immediately apparent. First, prices wander without a stable mean, ranging over more than an order of magnitude—from roughly 42 ¢/lb during the 2001–2002 bear market to a peak above 430 ¢/lb during the 2024–2025 rally. Second, the magnitude of daily returns varies substantially over time: the periods 1994–2000, 2011–2014, and 2024–2025 contain visibly larger moves than the calmer stretches in 2005–2010 and 2017–2020. Third, even within turbulent regions, the *sign* of returns shows no obvious pattern.

These observations correspond to three quantifiable stylized facts.

1) Stationarity: An Augmented Dickey-Fuller test [13] on P_t fails to reject the unit-root null hypothesis ($p \approx 0.56$), confirming that prices are non-stationary. The same test applied to r_t rejects the null overwhelmingly ($p < 10^{-30}$). Returns are therefore the natural object of analysis.

2) Fat-tailed return distribution: Daily returns have sample mean $\hat{\mu} = 1.88 \times 10^{-4}$ (approximately 4.7 % annualized at $\hat{\mu} \cdot 252$), sample standard deviation $\hat{\sigma} = 0.0237$ (approximately 37.6 % annualized at $\hat{\sigma}\sqrt{252}$), sample skewness 0.19, and *excess* kurtosis 6.65. The excess kurtosis is roughly an order of magnitude larger than would be expected from a Gaussian process; consistent with this, the daily series contains 31 moves of magnitude exceeding 10 %, where a Gaussian distribution calibrated to the sample variance would predict essentially zero such moves over the same sample size.

Fig. 5(a) plots the empirical density against maximum-likelihood Gaussian and Student's t fits, and Fig. 5(b) shows the corresponding Normal Q-Q plot. Both ends of the Q-Q plot curve sharply away from the diagonal, the signature of tails fatter than the Gaussian benchmark; the Student's t density captures both the high central peak and the heavy tails that the Gaussian cannot accommodate.

3) Volatility clustering: Fig. 6 plots the empirical autocorrelation function (ACF) of returns r_t and of their squared values r_t^2 at lags $1, \ldots, 40$. The two panels are visually distinct. The ACF of r_t [Fig. 6(a)] sits essentially within the 95 % confidence band at all lags, consistent with returns

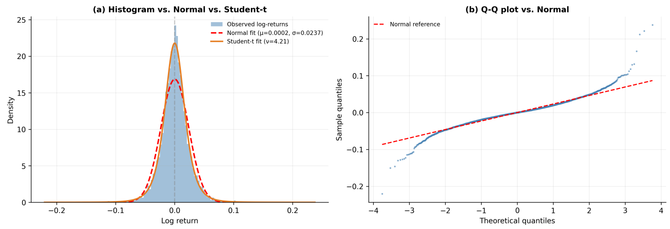

Fig. 5. Distributional diagnostics for daily log returns. (a) Empirical density (bars) versus fitted Gaussian (dashed) and Student's t (solid) densities. The Student's t density captures both the high central peak and the heavy tails; the Gaussian fits neither. (b) Normal Q-Q plot. Sharp departures from the diagonal at both ends confirm tails fatter than Gaussian.

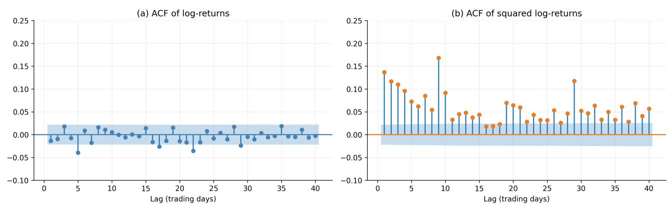

Fig. 6. Autocorrelation functions of (a) daily log returns r_t and (b) squared returns r_t^2, lags 1 through 40. Shaded regions are 95 % confidence bands under the null of zero autocorrelation. The contrast between the two panels is the empirical signature of volatility clustering. A Ljung-Box test [14] of zero autocorrelation through lag 20 yields $p \approx 0.006$ for r_t and $p < 10^{-200}$ for r_t^2.

themselves being approximately uncorrelated and consistent with Section I's finding that no point-forecast model could beat RWD. The ACF of r_t^2 [Fig. 6(b)] is large at lag 1 (≈ 0.14) and decays slowly, remaining well above the confidence band for at least 40 lags.

This contrast—near-zero autocorrelation in returns but strong, persistent autocorrelation in their magnitudes—is the empirical content of *volatility clustering*. It implies that while the next return's direction is essentially unforecastable, its expected *magnitude* can be predicted from recent magnitudes. Modeling this structure is the purpose of the GARCH framework.

C. Model Specification

Following [15], [16], we specify the daily return process as

$$r_t = \sigma_t z_t, \quad z_t \overset{\text{i.i.d.}}{\sim} t_\nu^*, \tag{1}$$

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2, \tag{2}$$

where t_ν^* denotes the standardized Student's t distribution with ν degrees of freedom and unit variance, and where $\omega > 0$, $\alpha, \beta \geq 0$, with $\alpha + \beta < 1$ for stationarity. The conditional variance σ_t^2 is a deterministic function of past returns and past variances: today's variance is a weighted average of a constant baseline (ω), the most recent squared shock ($\alpha\, r_{t-1}^2$), and the previous variance forecast ($\beta\, \sigma_{t-1}^2$).

This specification differs from the GARCH variant evaluated in Section I-C in three respects: the mean equation is fixed at zero rather than AR(1); the model is fit on log returns rather than raw prices; and the innovations are Student's t rather than Gaussian. All three changes are motivated by the stylized facts in Section II-B.

The Student's t innovation distribution generalizes the standard normal ($\nu \to \infty$) by introducing a single tail-heaviness parameter. Its inclusion is motivated empirically by the kurtosis result in Section II-B; mechanically, it permits the conditional distribution of r_t given $\mathcal{F}_{t-1}$ to assign meaningful probability to large-magnitude moves even after volatility clustering has been accounted for by (2).

Multi-step variance forecasting. For a forecast made at time T of return r_{T+h}, the GARCH(1,1) recursion (2) implies

$$\hat{\sigma}_{T+h|T}^2 = \omega \sum_{j=0}^{h-1} (\alpha+\beta)^j + (\alpha+\beta)^{h-1}\left(\alpha r_T^2 + \beta \sigma_T^2\right). \tag{3}$$

Under the working assumption that conditional returns are uncorrelated, the h-day cumulative log return $R_h = \sum_{j=1}^h r_{T+j}$ has conditional variance $\sum_{j=1}^h \hat{\sigma}_{T+j|T}^2$. Quantiles of R_h are obtained as

$$q_{R_h}(\tau) = \sqrt{\sum_{j=1}^h \hat{\sigma}_{T+j|T}^2} \cdot t_\nu^{*-1}(\tau),$$

and quantiles of the future price $P_{T+h} = P_T \exp(R_h)$ as $P_T \exp\big(q_{R_h}(\tau)\big)$.

Parameter estimation. Parameters $(\omega, \alpha, \beta, \nu)$ are estimated by maximum likelihood using the `arch` package [17]. On the full sample we obtain $\hat{\omega} = 0.099$, $\hat{\alpha} = 0.041$, $\hat{\beta} = 0.941$, and $\hat{\nu} = 5.51$ (after rescaling returns to percent). The persistence $\hat{\alpha} + \hat{\beta} = 0.982$ implies slow decay of variance shocks, consistent with the slow decay observed in the ACF of r_t^2. The implied unconditional volatility $\omega/(1-\alpha-\beta)$ corresponds to 37.7 % annualized, matching the full-sample sample standard deviation to within rounding.

D. Backtest Methodology

To enable direct comparison with Section I, the walk-forward backtest of Section II uses the same 60 forecast origins spaced 110 days apart and the same 63-day forecast horizon. The training window, however, differs: Section II uses an *expanding* window—all available log returns prior to each forecast origin—rather than the fixed-length rolling window used in Section I. The choice is motivated by the sample-sensitivity of the Student's t tail parameter ν. Reliable estimation of ν benefits from observing many tail events; expanding the training window across all available history gives ν exposure to the high-volatility regimes of 1994–2000, 2011–2014, and 2024–2025 that any single 1 536-day window would only partially span. At each origin T_k (for $k = 1, \ldots, 60$), we

1) refit the GARCH(1,1)+t model on all log returns up to T_k, i.e. $\{r_2, \ldots, r_{T_k}\}$;
2) generate the predictive distribution of R_h for $h = 1, \ldots, 63$ using (3);
3) translate distribution quantiles into price quantiles via $P_{T_k} \exp(\cdot)$;
4) store the predicted quantiles and the realized price path $\{P_{T_k+1}, \ldots, P_{T_k+63}\}$.

TABLE II
POOLED EMPIRICAL COVERAGE OF GARCH(1,1)+t PREDICTION INTERVALS ACROSS ALL 3,780 FORECAST POINTS. DEVIATION IS THE DIFFERENCE BETWEEN EMPIRICAL AND NOMINAL COVERAGE IN PERCENTAGE POINTS.

Nominal level	Empirical coverage	Deviation
50 %	49.26 %	-0.74
80 %	79.50 %	-0.50
90 %	91.03 %	$+1.03$
95 %	97.30 %	$+2.30$
99 %	99.84 %	$+0.84$

Evaluation. Calibration is the primary evaluation criterion [18], [19]. We compute *empirical coverage* of central prediction intervals at nominal levels $\{50, 80, 90, 95, 99\}$ %, defined as the fraction of realized observations falling between the model's $\frac{1-\tau}{2}$ and $\frac{1+\tau}{2}$ predicted quantiles. A well-calibrated forecast has empirical coverage close to the nominal level at every level evaluated.

E. Results

Table II reports pooled empirical coverage of the GARCH(1,1)+t model across all $60 \times 63 = 3{,}780$ forecast points. Empirical coverage closely tracks nominal coverage at every level evaluated, with absolute deviations ranging from 0.50 pp at the 80 % level to 2.30 pp at the 95 % level. The model under-covers slightly at the lower levels (50 % and 80 %) and over-covers slightly at the higher levels (90 %, 95 %, and 99 %); the mean absolute deviation across the five levels is 1.08 pp.

Fig. 7 disaggregates coverage by forecast horizon at the 80 % and 95 % levels. Two patterns are visible. At short and intermediate horizons (roughly $h \leq 30$), empirical coverage runs below the nominal target at both levels; at longer horizons ($h > 40$), coverage drifts above target. These two effects partially offset when pooled, yielding the small mixed deviations reported in Table II: net under-coverage of under 1 pp at the 50 % and 80 % levels, net over-coverage of 1–2 pp at the 90 %, 95 %, and 99 % levels.

Fig. 8 illustrates GARCH(1,1)+t forecasts at six origins spanning the breadth of price regimes in the backtest: calm periods (Dec 2001, Dec 2018), moderate-volatility periods (May 2016, May 2022), the 2011 rally (Mar 2011), and the 2024–2025 rally (May 2025). The figure makes visible the central operational feature of the model: predictive-interval widths contract during calm regimes and expand during turbulent ones, in proportion to conditional volatility at each origin. The 95 % band at Origin 5 (Dec 2001) is narrow, reflecting the low-volatility conditions in late 2001; the corresponding band at Origin 59 (May 2025) is roughly an order of magnitude wider in absolute terms, reflecting the elevated volatility of the 2024–2025 rally. All six realized paths fall within their 95 % prediction intervals, with Origin 59 approaching the lower 80 % boundary but remaining inside the 95 % band.

F. Discussion

1) Why empirical coverage tracks nominal closely: The pooled deviations in Table II are small in absolute terms—under 2.5 pp at every level evaluated—and mixed in sign across the five nominal levels. Two observations frame this result. First, the effective sample size is much smaller than the $3{,}780$ raw forecast points: returns within a single 63-day forecast trajectory are highly dependent, so the effective number of independent trials is closer to the 60 origins. Under that effective sample size, the observed deviations sit within one standard error of nominal calibration at every level—they are not statistically distinguishable from perfect calibration. Second, the mixed sign of the deviations (under-coverage at the lower nominal levels, over-coverage at the higher ones) is consistent with the empirical innovation distribution being slightly more peaked but slightly thinner-tailed than the fitted Student's t. No systematic bias is visible in the data.

2) Regime adaptability and the limits of parametric tails: Fig. 8 demonstrates the central operational strength of GARCH(1,1)+t: predictive-interval widths scale with local volatility, narrowing during calm periods and widening during turbulent ones. Parametric tail modeling has limits, however. Even with the heavier-tailed Student's t innovations, sufficiently extreme tail events—those outside any reasonable parametric distribution—would require non-parametric extensions (e.g. extreme value theory in the tails), regime-switching variance dynamics, or sources of information external to the price series itself (e.g. weather, inventory data) to model directly.

3) Limitations: We model a single price series under a parametric GARCH(1,1) specification. The literature contains numerous extensions—asymmetric variance dynamics (GJR-GARCH, EGARCH), long-memory variants (FIGARCH), and stochastic volatility models—that may further improve calibration; in tests on equity indices these typically yield small marginal gains relative to a well-specified GARCH(1,1) [20], but their performance on commodities is less extensively documented. Our backtest also fits all parameters by maximum likelihood; Bayesian approaches that quantify parameter uncertainty would yield wider predictive intervals at the cost of additional computational complexity.

G. Operational Implementation

The GARCH(1,1)+t specification validated in Section II-E is implemented as an automated daily forecasting pipeline for 19 Coffee Company. Each trading day after market close, the pipeline appends the latest settlement price to the historical series, refits the model on all available history (matching the expanding-window design of the calibration backtest), and produces an updated 63-day fan chart with 80 % and 95 % prediction intervals. Price updates after the ICE Report 293 archive cutoff are pulled from a real-time exchange data feed; the historical training data underlying the backtest in Section II-E is unchanged. Fig. 9 shows a representative output. The continuously-refreshed fan chart provides 19 Coffee with a current-state-aware estimate of plausible 3-month price

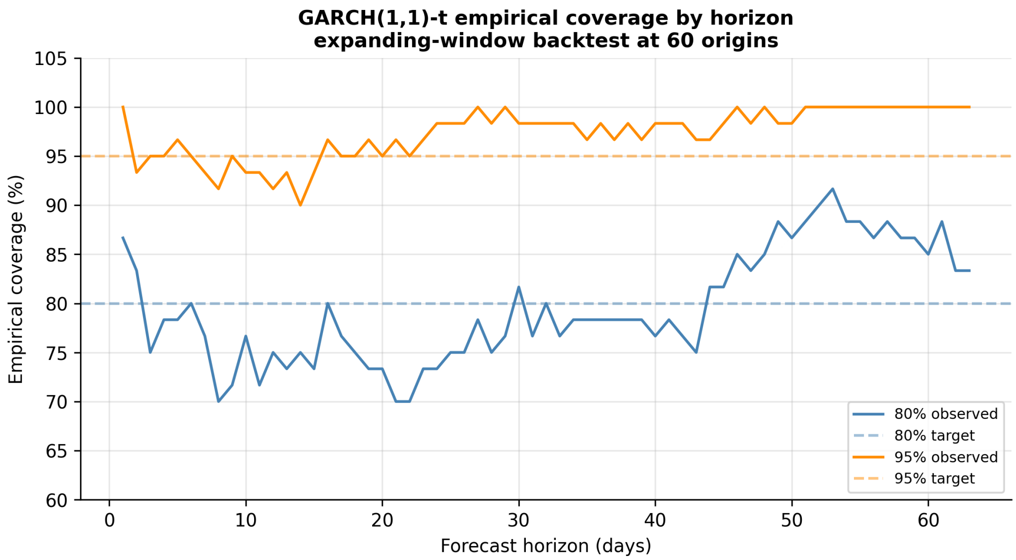

Fig. 7. Empirical coverage of GARCH(1,1)+t prediction intervals at the 80 % and 95 % levels as a function of forecast horizon, averaged across the 60 origins. Dashed lines show the nominal target rates. Empirical coverage tracks the nominal rates within a small band at all horizons evaluated.

trajectories, supporting procurement and budgeting decisions on an ongoing basis rather than at a single fixed-date forecast.

III. Conclusion and Future Work

Section I of this paper found that no point-forecast model in a deliberately diverse ten-model suite could materially improve on a random-walk-with-drift baseline for daily ICE Coffee C futures prices, with nine of ten models jointly indistinguishable from the best at $\alpha = 0.10$ on a Model Confidence Set test. Section II demonstrates that this result is not evidence of an unstructured series. When the forecasting target is changed from a point estimate to a predictive distribution, the persistent volatility clustering and fat-tailed innovations in coffee returns become exploitable: a GARCH(1,1) model with Student's t innovations, fit on expanding training windows and evaluated at 60 origins, produces prediction intervals whose empirical coverage tracks nominal coverage within a few percentage points at every level evaluated, with interval widths that adapt to local volatility across diverse historical regimes.

For 19 Coffee Company, this means the deliverable is not a price oracle but an honest uncertainty band: at any given trading day, the GARCH-t model produces an 80 % interval whose realized coverage tracks its nominal rate closely, and which is meaningfully tighter during calm volatility regimes and wider during turbulent ones. Such an interval supports concrete decisions—when to lock in a purchase, how much budget reserve to maintain, when to escalate procurement discussions—in ways that a point forecast cannot.

The methodological lesson generalizes beyond coffee: scoring a forecast by MAE answers a different question from scoring it by interval coverage, and the right scoring rule depends on the downstream decision the forecast informs. When decisions involve risk management, position sizing, derivative pricing, or any task sensitive to the spread of possible outcomes, density forecasts are strictly more informative and should be evaluated as such.

Several extensions suggest themselves. An out-of-sample comparison against asymmetric GARCH variants and stochastic-volatility models would clarify whether further calibration gains are available. Multi-asset joint modeling—in particular against the Brazilian Real and Arabica/Robusta spreads—may exploit cross-market information unavailable to the univariate GARCH framework. Finally, augmenting the parametric tail model with extreme-value methods would provide additional robustness during extreme regimes.

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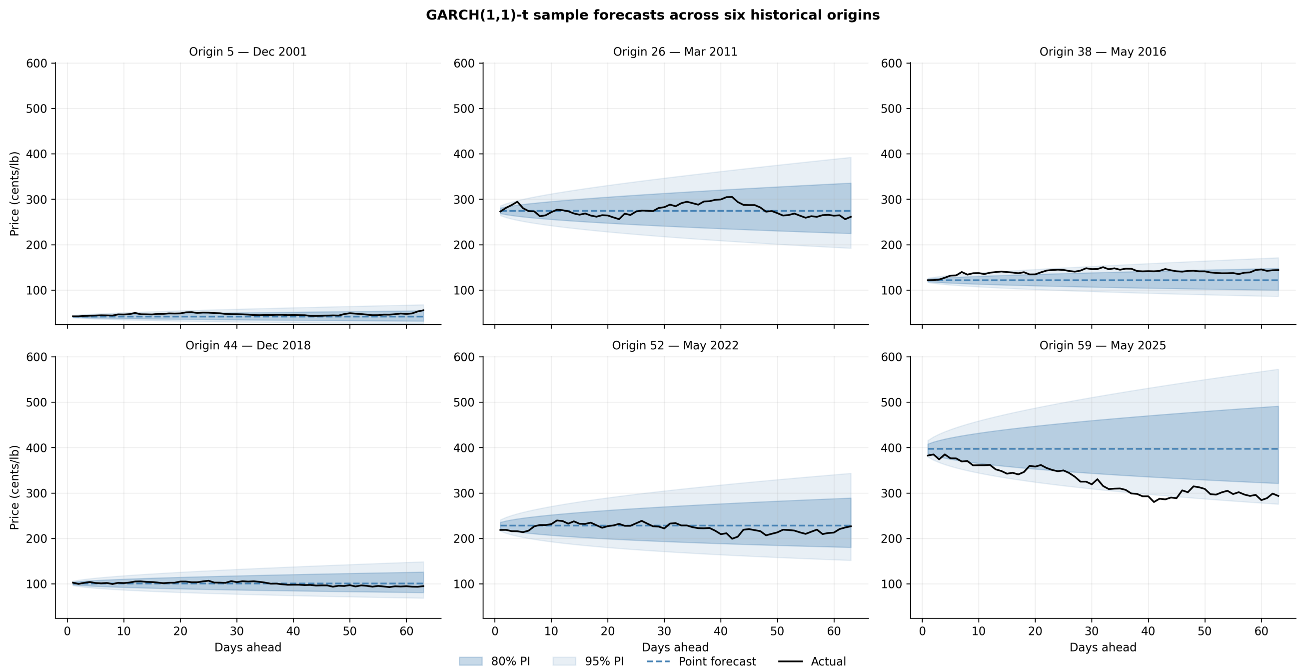

Fig. 8. GARCH(1,1)+t sample forecasts at six historical origins from the 60-origin backtest, spanning calm regimes (Dec 2001, Dec 2018), moderate volatility (May 2016, May 2022), and major rallies (Mar 2011, May 2025). Each panel shows the point forecast (dashed), 80 % and 95 % prediction intervals (shaded), and the realized price path (solid). Interval widths adapt to local volatility: tight during calm periods, wide during turbulent ones.

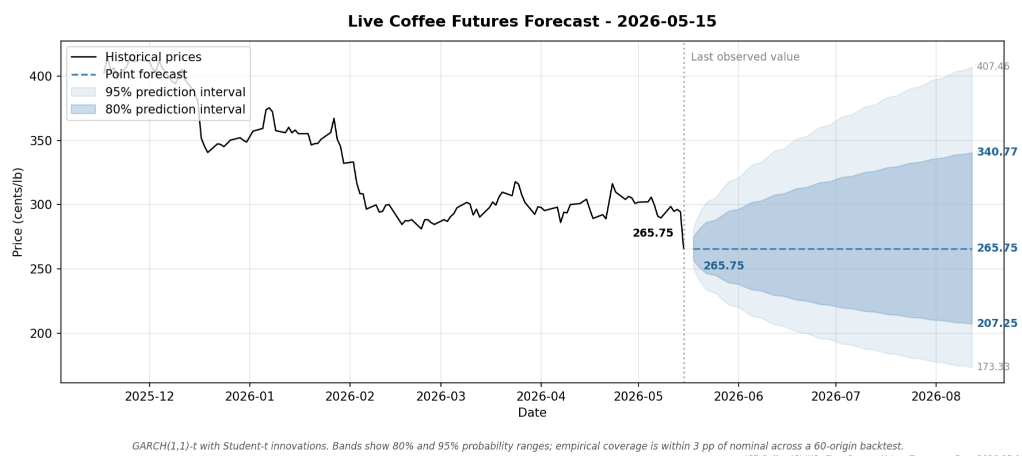

Fig. 9. Sample output of the operational forecasting pipeline, generated on May 13, 2026. Historical settlement prices (solid) are followed by the 63-day point forecast (dashed) and the 80 % and 95 % prediction intervals (shaded). The point forecast remains approximately flat at the most recent settlement, consistent with the near-zero conditional mean of returns; uncertainty grows with horizon according to the model's conditional variance forecast.

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